CS2060 Programming with C - Ch. 3 Assignment

1 Instructions

1. Submit this assignment on Canvas.
2. Solve each question in a separate (.c) file.
3. Print screen of the execution of each question (as evidence that your code is working).
4. It is highly recommended to run your code (for each question) on the test cases and print screen the results each time.
5. For each question, collect the .c file and its print screens into a folder.
6. Create a folder for the whole assignment.
7. Place all question folders into the assignment folder.
8. Compress the assignment folder into one zipped folder.
9. Upload this zipped folder to Canvas.
10. It is your own responsibility to check that you uploaded the assignment correctly (you can open the submission web page on Canvas and download your submission to check everything is OK).
11. If the Canvas is down and the deadline is too close, you can send the assignment via email to me and the TA with a print screen of the Canvas page that is down (as evidence that the Canvas website is down).

2 Questions

1. Write a C program to perform the following:
   - Asking the user to enter the number of values (positive integer).
   - Determining and displaying the maximum and the second maximum of them.

- **Hint:** you can use: while loop, if statement, and if...else statement. Do not use: for loop, do...while loop, and switch statement.

```
Please, enter the number of values (positive integer): 0
Invalid input.
```

```
Please, enter the number of values (positive integer): -1
Invalid input.
```

```
Please, enter the number of values (positive integer): 5
Please, enter an integer number: 1
Please, enter an integer number: 1
Please, enter an integer number: 1
Please, enter an integer number: 1
Please, enter an integer number: 1
The maximum is: 1
There is no second maximum.
```

```
Please, enter the number of values (positive integer): 6
Please, enter an integer number: 5
Please, enter an integer number: 6
Please, enter an integer number: 9
Please, enter an integer number: 8
Please, enter an integer number: 9
Please, enter an integer number: -1
The maximum is: 9
The second maximum is 8
```

2. Write a C program to perform the following:
   - Asking the user to enter the number of lines (positive integer).
   - Using iteration to print the following table of values.
   - **Hint:** you can use: while loop, if statement, and if...else statement. Do not use: for loop, do...while loop, and switch statement.

```
Please, enter the number of lines (positive integer): 0
Invalid input.
```

```
Please, enter the number of lines (positive integer): -1
Invalid input.
```

```
Please, enter the number of lines (positive
   integer): 5
```

```
Number     Number^2     Number^3
     1            1            1
     2            4            8
     3            9           27
     4           16           64
     5           25          125
```

3. Write a C program to perform the following:

   - Getting a non-negative integer number from the user using the keyboard.
   - Printing the sum of the digits of that number.
   - **Hint:** you can use: while loop, if statement, and if...else statement. Do not use: for loop, do...while loop, and switch statement.
   - **Hint:** use the division and remainder operators to separate the number into its individual digits.

```
Please, enter a number (non-negative integer): -1
Invalid input.
```

```
Please, enter a number (non-negative integer): 0
The sum of digits is: 0
```

```
Please, enter a number (non-negative integer): 123
The sum of digits is: 6
```

```
Please, enter a number (non-negative integer): 606
The sum of digits is: 12
```

```
Please, enter a number (non-negative integer): 500
The sum of digits is: 5
```